

Lecture 3 Flow Control and Sequence Programming

- Control principles
- Combinatorial control versus Flow control
- State diagram
- Flow Control with Sequential Function Chart (SFC)
- Example washing machine

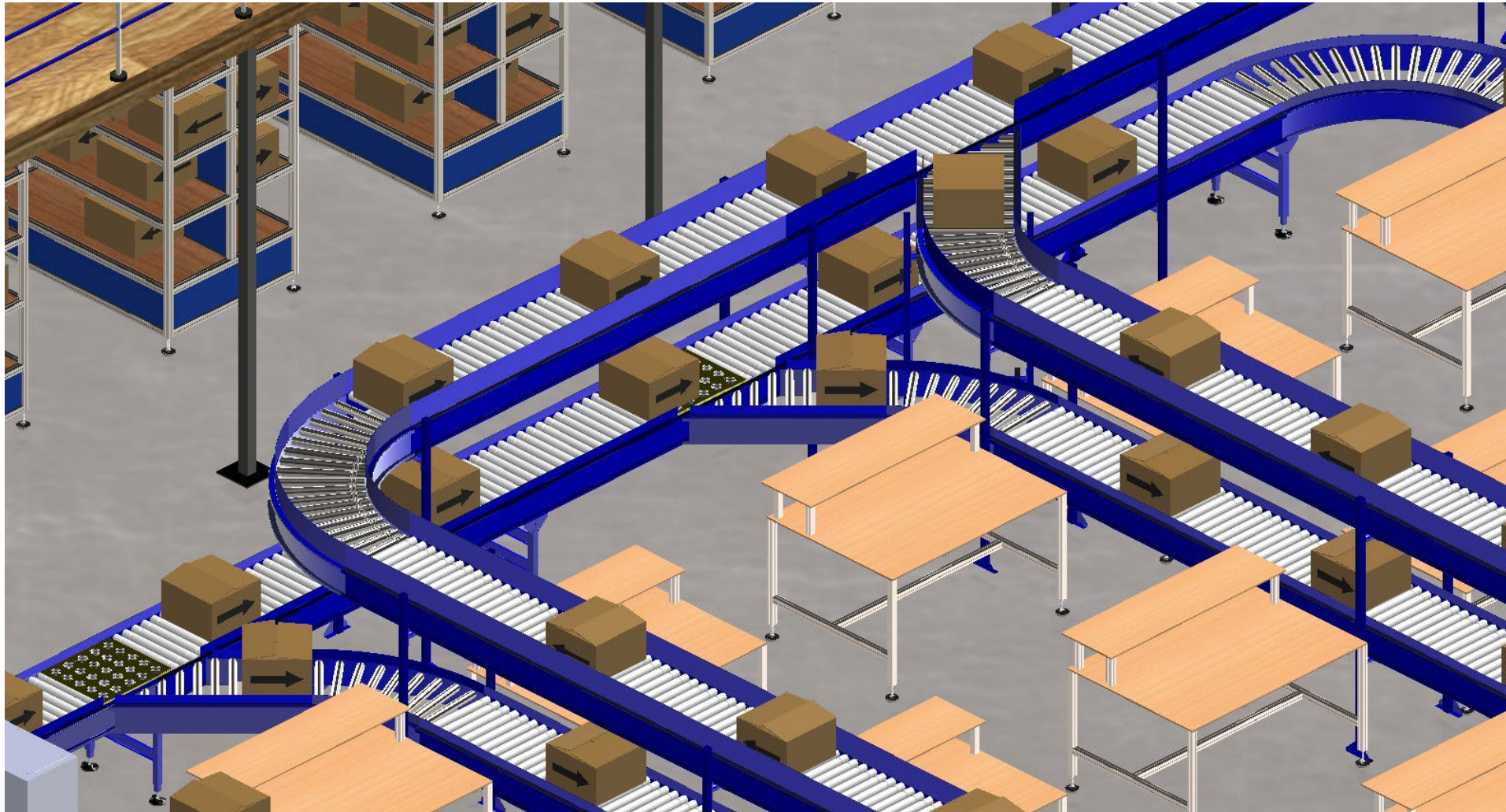
Classification of control principles

- Time Control
 - Time bases start of tasks
 - Few machines are fully Time-Controlled
- Combinatoric Control (Interlock programming)
 - The principle we have used so far in this course
 - Logic combinatoric of each output that determinates the operations
 - Action depends directly on input change
 - Provides complex and flexible programs
 - Suitable for unpredictable processes (logic based on sensors)
- Flow Control (Sequence Programming)
 - When the operations occur in predictable sequences
 - Action depends on input change and history (present state)
 - Suitable for predictable processes

When to use Combinatorial Logic (extended with independent memories)

- When **High Flexibility** is needed and when a sequence is not predictable
 - Conveyors
 - Elevators
 - Human interaction
- Drawbacks
 - Requires much of the programmer
 - Hard to find out what the machine really does by reading the program

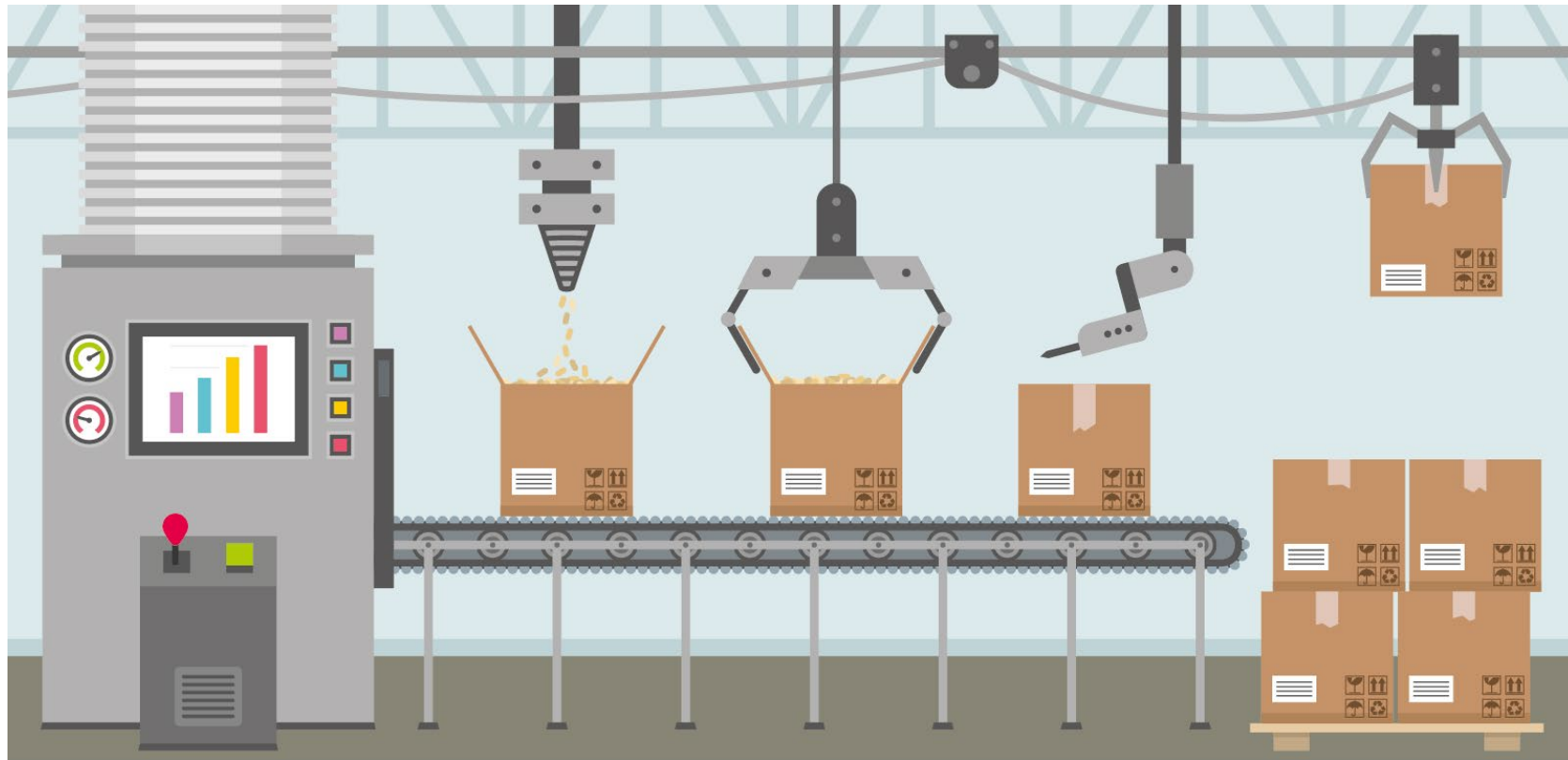
Conveyor system = Combinatoric control



Sequence or Flow controlling

- Used for predefined sequential operations
 - Parallel and Alternative sequences are possible
- Easy to understand what the machine does by reading the program
- Example:
 - Washing machines
 - Assembly processes
 - Packaging machines
- Drawback
 - Inflexible
 - Bad response to unpredictable changes

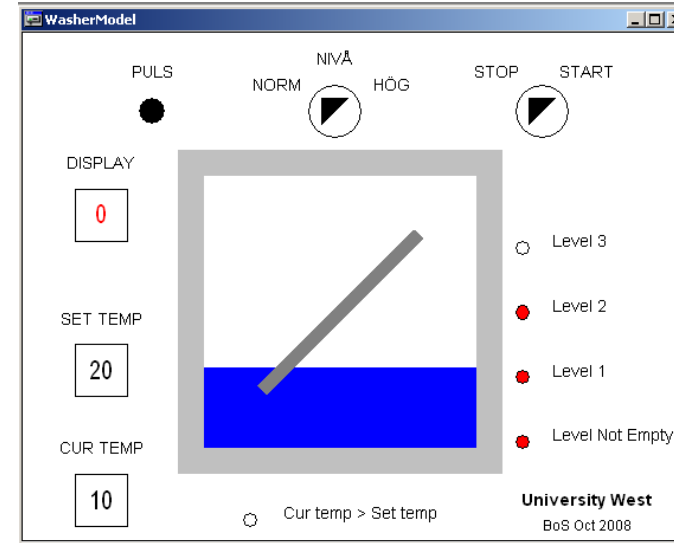
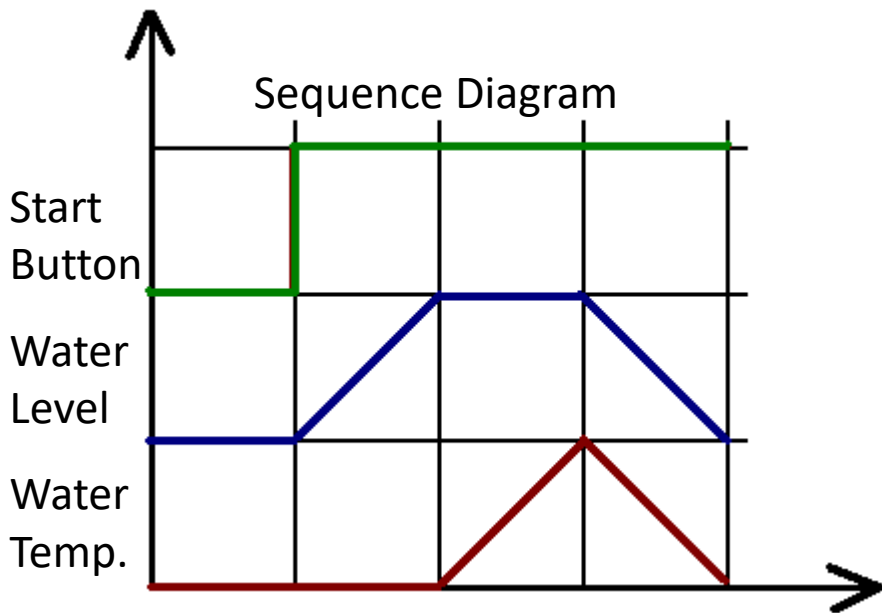
Packing machine = Sequence control?



Structured design – methods

- Logical functional expression
- Time path diagram
- Sequence diagram
- Flowchart
- State diagram

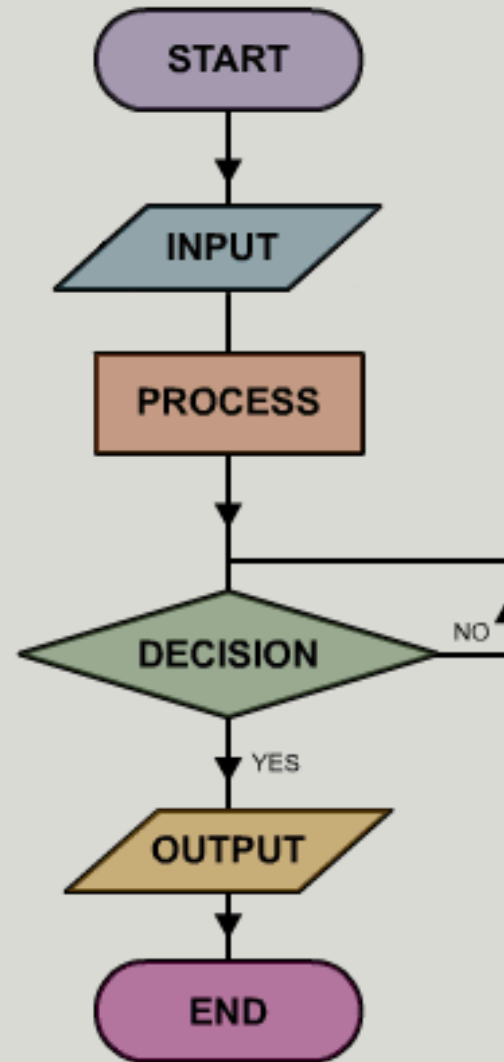
Example Sequence



1. Wait on Start Button
2. Fill water to level 2
3. Heat the water
4. Empty water

Flowchart symbols

System flowchart symbols



All flowcharts begin with the **START** symbol. This shape is called a terminator.

INPUTS, such as materials or components,
eg Printed Circuit Board (PCB)

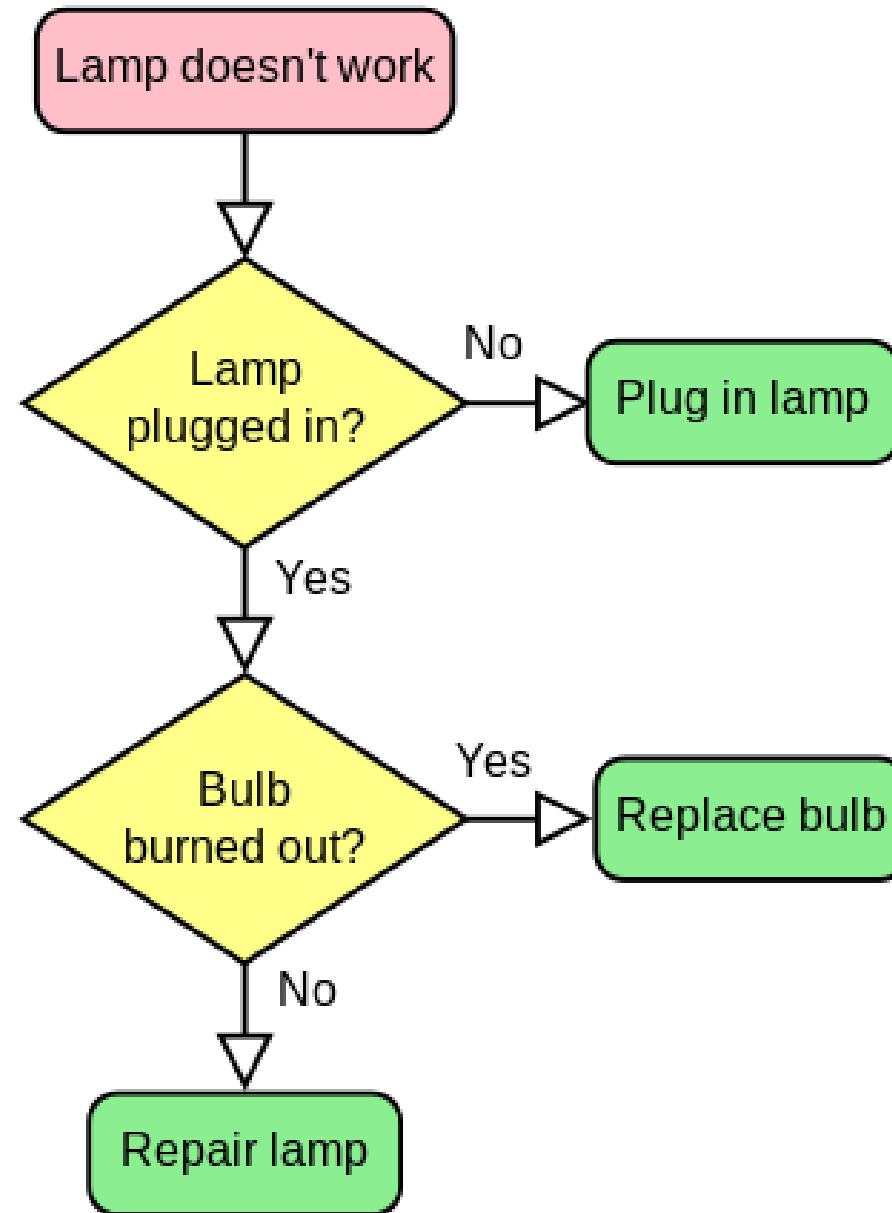
PROCESSES, such as activities or tasks, are sometimes used to link to a subroutine (another flowchart) with more detailed steps, eg drill Printed Circuit Board (PCB)

The **DECISION** symbol checks a condition before carrying on, eg is the drilling accurate?

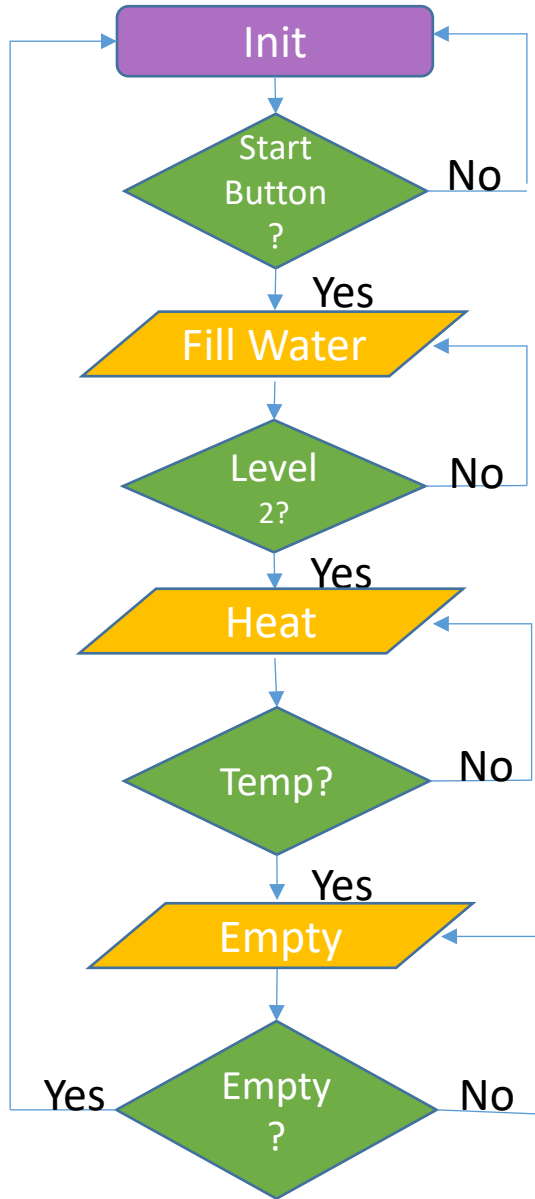
OUTPUTS, eg Printed Circuit Board (PCB) with holes drilled.

All flowcharts end with the **END** symbol. This shape is called a terminator.

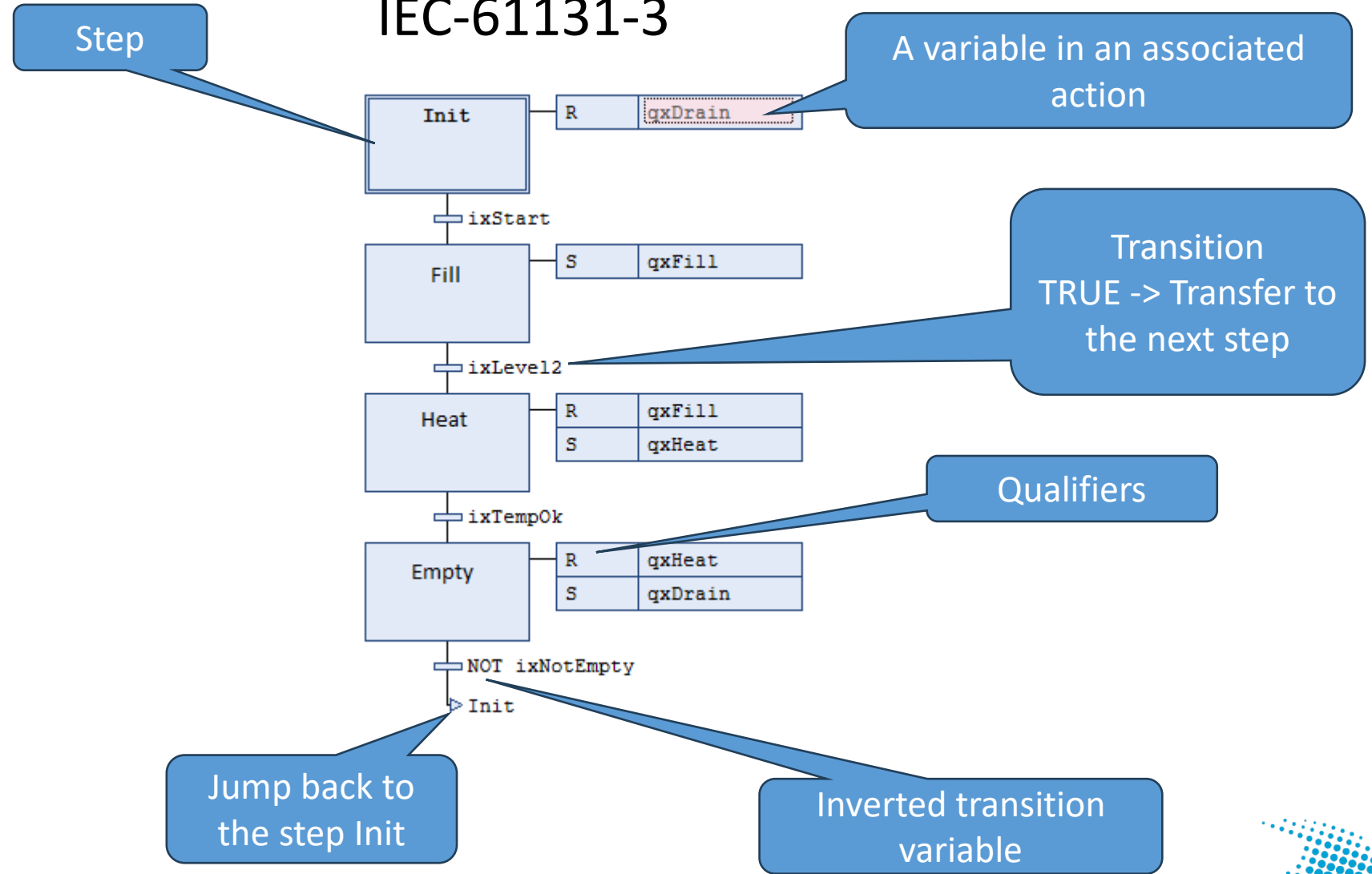
Flowchart – example



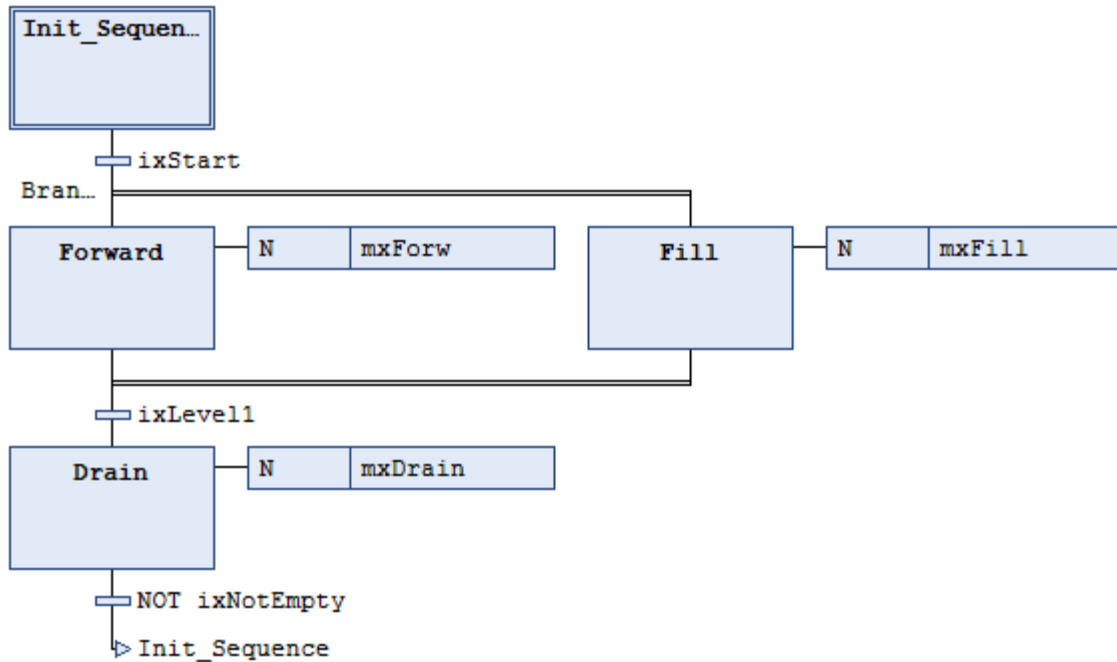
Flow Chart



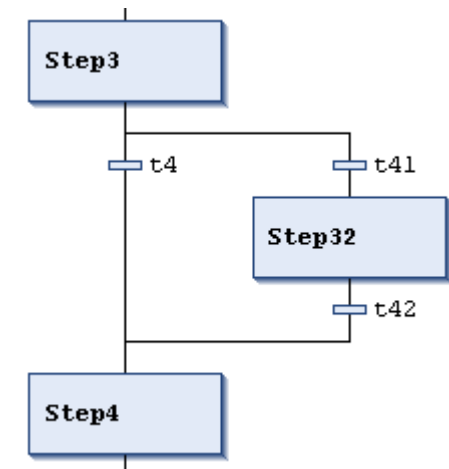
Sequence Function Chart (SFC) IEC-61131-3



Double lines are parallel steps



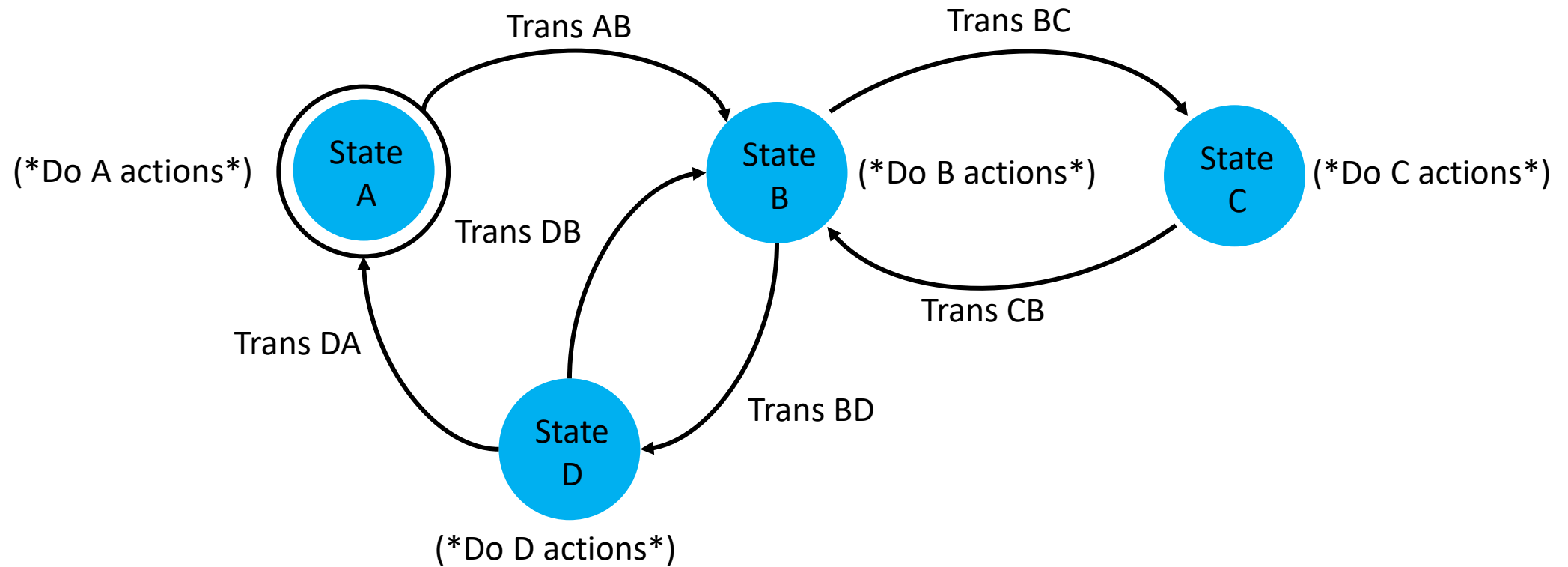
Single lines are
alternative steps



State diagram

- Same information as flowchart, but more compact
- Useable for both combinatorial and sequential system
- Illustrate
 - all possible states
 - which state follow one another
 - which condition must be fulfilled to change state

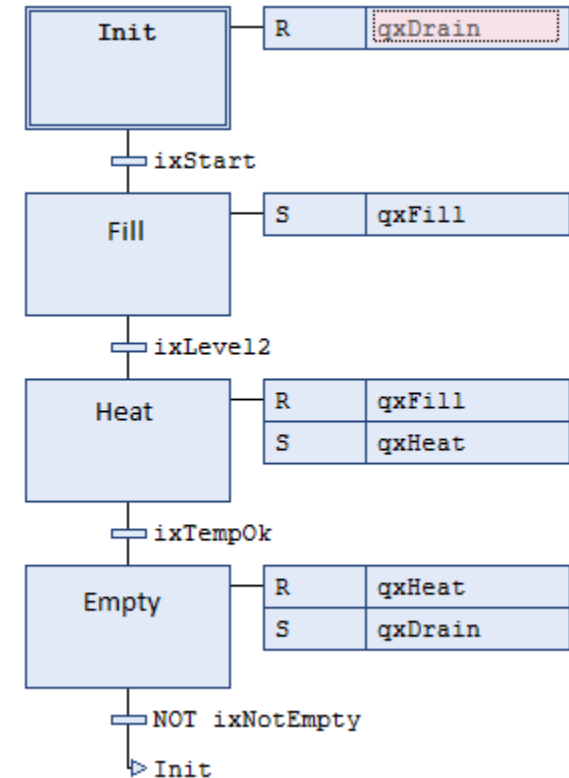
State diagram – example



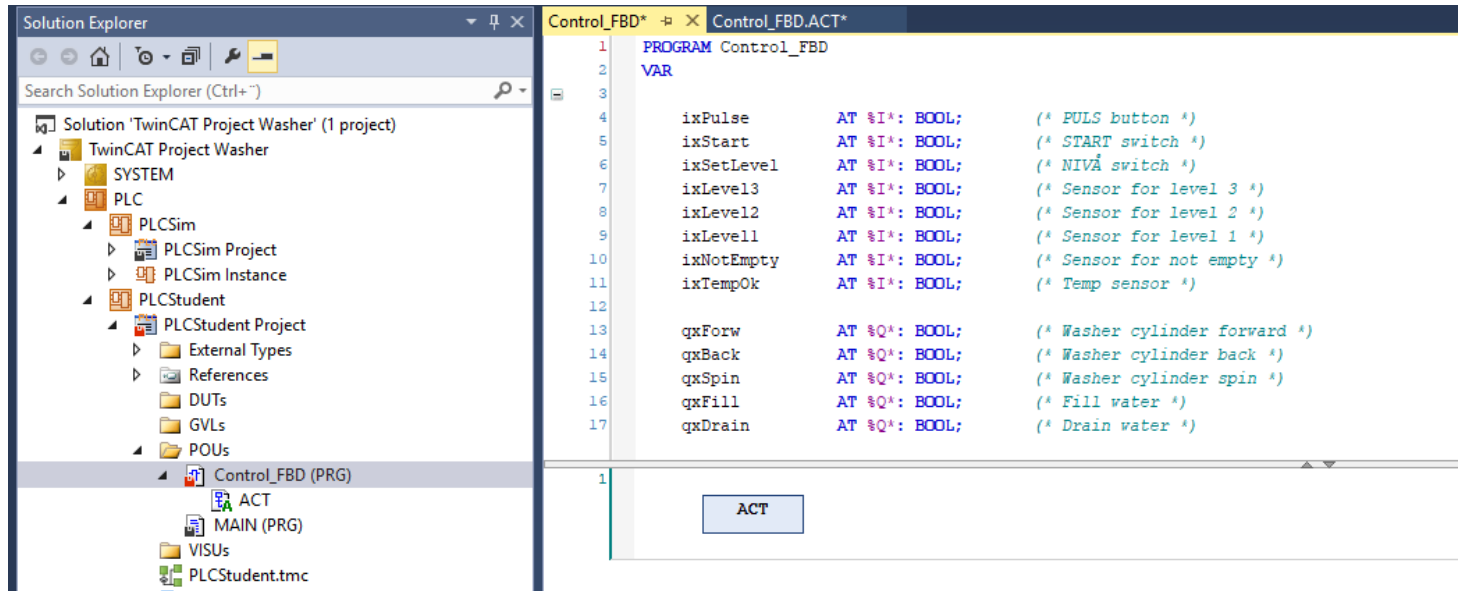
- Deterministic behavior requires only one satisfied transition at a time from a state

The washing machine in SFC

- Start Washing machine project in TwinCAT
- Create a new POU with the language SFC
- Declare in and outputs in the header
- Build up the SFC sequence



Add SFC as an action in the POU Control_FBD

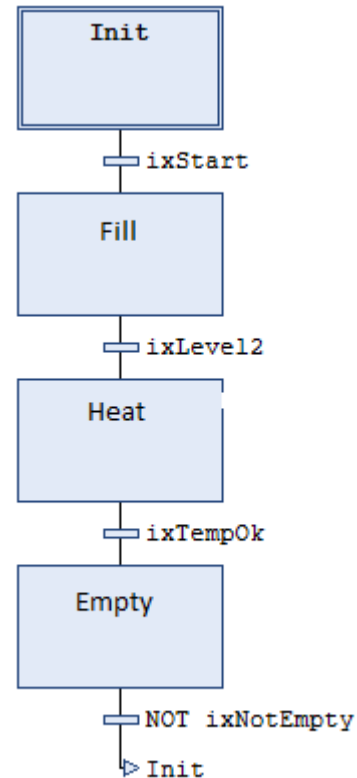


Right-click on
Control_FBD
Select language SFC
Give the action a name
Call the action within
the FBD
Save and activate the
configuration

- This arrangement makes it possible to combine FBD and SFC and reuse the variables

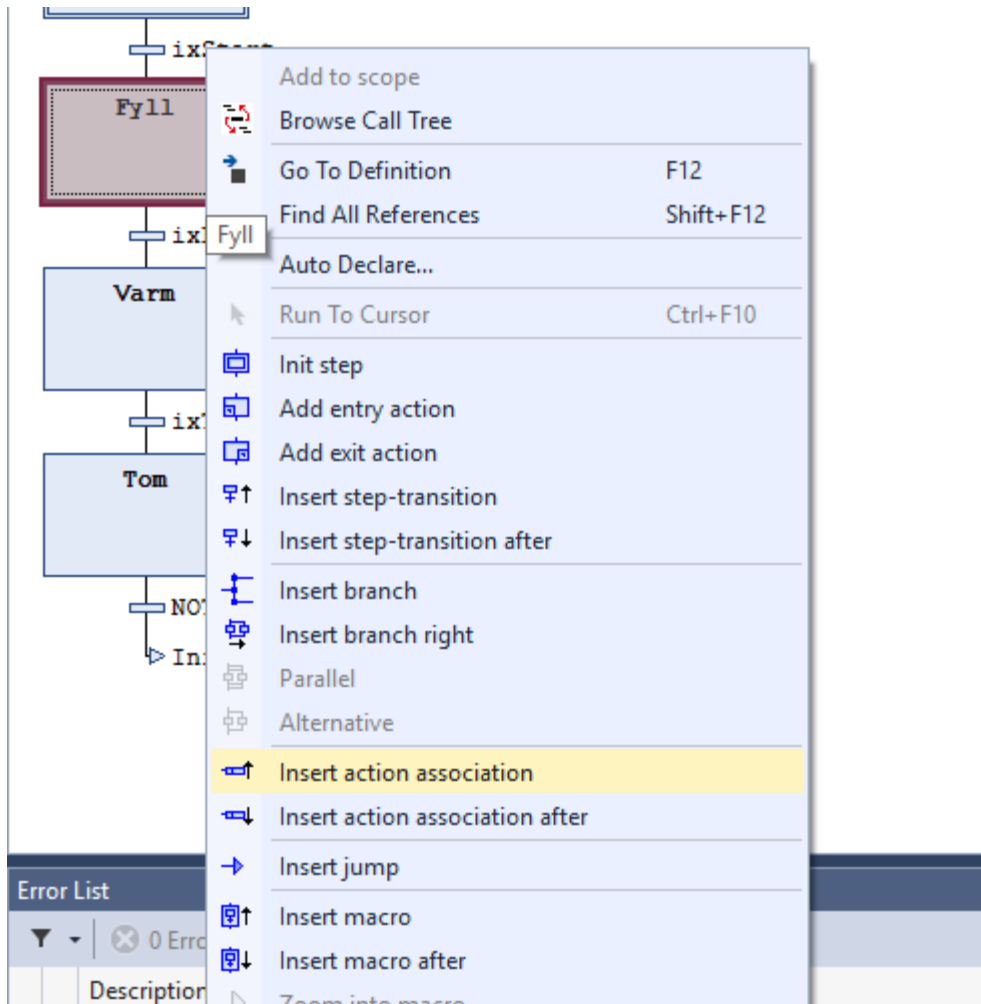
Give the steps proper names

Insert the Transition variables



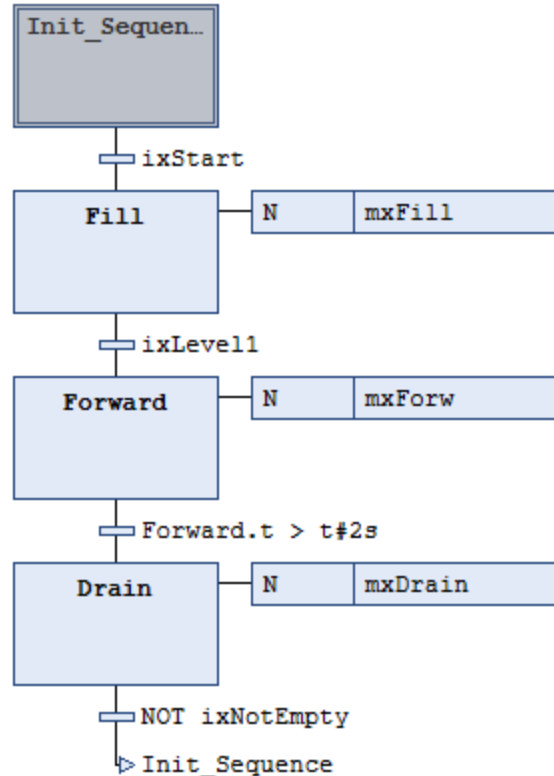
- Use drag and drop from the toolbox
- Or right-click to add steps and transitions

Action Association



- Right-click on a step
- Do “inset action association”

Qualifiers of the associated actions to variables



N	Non-stored	The associated variable is TRUE as long as the step is active else FALSE
R	Reset (Stored)	The Associated variable is set to FALSE
S	Set (Stored)	The Associated variable is set to TRUE

The washing machine sequence by a CASE in ST

CASE miSequence OF

0: //Initiate

qxDrain := FALSE;

IF ixStart THEN miSequence := 10; END_IF

10: //Fill

qxFill := TRUE;

IF ixLevel2 THEN miSequence := 20; END_IF

20: //Heat

qxFill := FALSE;

qxHeat := TRUE;

IF ixTempOk THEN miSequence := 30; END_IF

30: //Emty

qxHeat := FALSE;

qxDrain := TRUE;

IF NOT ixNotEmpty THEN miSequence := 0; END_IF

END_CASE

VAR

miSequence: INT;

END_VAR

Chapter 4 in Programmable Logic Controller

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- Read through Chapter 4
- 4.5 Sequential Design
- Test Problems 4.10